

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **data-driven simulation models** using **Python** and **R statistical methods** to optimise rail network **capacity planning** and **timetabling performance**, reducing modelled delays by **18%** and improving passenger throughput forecasts across multi-disciplinary feasibility studies.
- Led **digital engineering** and **safety assurance** initiatives by applying **Power BI dashboards** and **operational reporting** to track reliability metrics, incident patterns, and asset performance in real-time across interconnected rail infrastructure systems, enabling data-driven decisions for **10+ improvement proposals**.
- Engineered **REST API integrations** and **system interconnections** between rail operational platforms using **Java Spring Boot**, **PostgreSQL**, and **AWS (EC2, Lambda)** to synchronise timetable, capacity, and maintenance data across **5 mission-critical enterprise systems**, achieving **99.8% uptime** and eliminating manual data reconciliation.
- Executed **change-control management** and **UAT coordination** for rail infrastructure upgrades and system configuration updates, collaborating with cross-functional teams including engineering stakeholders, safety assurance leads, and delivery partners to ensure **zero post-implementation incidents** across **12 major deployments**.
- Delivered **concept and feasibility studies** using **data modelling** and **business intelligence** frameworks (**Tableau**, **DAX**, **Power Query ETL**) to quantify network optimisation opportunities, decarbonisation pathways, and passenger experience improvements, influencing **€2.3M in approved infrastructure investment**.
- Coached junior engineers and mentored **4 graduate-level colleagues** on best practices in **systems engineering**, **digital simulation methodologies**, and **operational modelling** through structured training, code reviews, and knowledge-sharing sessions, advancing team delivery capability and professional chartership progression.

SKILLS

Rail Engineering & Operations – Rail operations & planning, capacity modelling, timetabling optimisation, operational simulation, safety assurance, systems engineering, asset reliability, decarbonisation.

Data-Driven Methods & Simulation – Python simulation models, R statistical analysis, hypothesis testing, time-series forecasting, network optimisation, digital engineering, scenario modelling.

Business Intelligence & Digital Systems – Power BI dashboards, Tableau interactive visualisations, DAX measures, Power Query ETL, data modelling, data warehousing, operational reporting, KPI tracking.

Backend Engineering & Cloud Infrastructure – Java Spring Boot, REST APIs, system integrations, PostgreSQL, MySQL, Oracle, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins CI/CD.

Delivery, Change Management & Stakeholder Engagement – Concept and feasibility studies, detailed design assurance, UAT coordination, change-control management, cross-functional collaboration, proposal development, thought leadership.

Professional Development – Coaching and mentoring, knowledge sharing, best-practice documentation, Agile/Scrum, Jira, Confluence, professional accreditation pathways.

Certifications & Qualifications – MSc Data Analytics (National College of Ireland), Bachelor of Engineering Computer Science, Full Stack Java Spring Boot & Angular, Web Development (Coursera), Python Programming (Coursera), IELTS Academic Band 7/9.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Rail Network Capacity & Timetabling Optimisation Platform (HCL Technologies)

- Architected an **integrated digital engineering platform** combining **Python simulation models** with **Power BI operational dashboards** to forecast rail network capacity constraints, model timetabling scenarios, and quantify decarbonisation opportunities across **150+ stations** and **3 freight corridors**.
- Implemented **R statistical methods** (regression analysis, time-series forecasting, Monte Carlo simulation) to analyse passenger demand patterns and service reliability, enabling **25% improvement in timetable adherence** predictions and **18% reduction in modelled peak-hour delays**.
- Deployed **REST API integrations** using **Java Spring Boot** to synchronise real-time operational data from **5 interconnected systems** (signalling, asset management, scheduling, safety assurance, maintenance) into a unified **PostgreSQL data warehouse**, supporting **12+ feasibility studies** across concept and detailed design phases.
- Designed **Power BI dashboards** with **DAX-driven KPI measures** (punctuality, capacity utilisation, reliability metrics) and **Power Query ETL pipelines** to transform raw operational telemetry into executive-facing strategic reports, accelerating decision-making for **€2.3M infrastructure investment approval**.
- Executed **safety assurance** validation and **change-control management** across platform releases, coordinating **UAT with 8 multi-disciplinary teams** (engineering, operations, safety leads, delivery partners) to achieve **zero post-implementation issues** and **99.8% production uptime** over 18-month deployment.

Rails Operations Dashboard & Incident Reliability Analytics (HCL Technologies)

- Engineered an **operational monitoring and safety assurance dashboard** using **Tableau and Power BI** to visualise real-time rail asset performance, incident trends, and reliability metrics across **6 regional networks**, enabling proactive **systems engineering** interventions and reducing unplanned downtime by **22%**.
- Built **data pipeline automation** using **Python ETL scripts** and **AWS Lambda** to ingest operational data from legacy **PostgreSQL and Oracle databases**, cleanse with **pandas and NumPy**, and publish curated datasets to **Power BI** for **real-time stakeholder reporting** across safety, capacity, and decarbonisation KPIs.
- Collaborated with cross-functional **multi-disciplinary teams** (rail operations leads, safety assurance engineers, asset managers) to translate business requirements into data-driven recommendations, supporting **concept feasibility studies** and **detailed design assurance** for **5 major infrastructure upgrades**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)